

SaaS Product Analytics

Customer Churn Analysis Report

Telco Customer Churn Dataset | 7,043 Customers | 21 Features

Tools: Python | SQL | XGBoost | LightGBM | SHAP | Power BI

26.54%

Churn Rate

0.848

Model AUC-ROC

\$1.67M

Revenue Lost / yr

282

High-Risk Customers

This report presents a full end-to-end product analytics study on customer churn for a SaaS telecom provider. It covers data preparation, exploratory analysis, SQL-based business queries, predictive modeling, model explainability, and a three-page Power BI dashboard delivering actionable retention insights.

1. Executive Summary

Customer churn is one of the highest-leverage problems a SaaS business can address. Acquiring a new customer costs 5-7x more than retaining an existing one. This analysis reveals that 26.54% of the provider's 7,043 customers have churned, resulting in \$139K in monthly recurring revenue loss and \$1.67M annualized. Three segments account for the majority of this loss: month-to-month contract holders, fiber optic internet subscribers, and customers in their first 12 months of tenure.

Key finding: A single profile — month-to-month contract, fiber optic internet, electronic check payment, first-year customer — accounts for the highest concentration of churned customers and represents the primary target for retention intervention.

2. Dataset & Data Preparation

The Telco Customer Churn dataset (IBM Watson Analytics) contains 7,043 customer records with 21 attributes covering demographics, subscribed services, account information, and churn status.

Data quality issues resolved

Issue	Column	Resolution
Wrong dtype (string)	TotalCharges	Converted to float64 using <code>pd.to_numeric()</code>
11 null values	TotalCharges	Filled with 0 (new customers, tenure=0)
Binary as string	Churn	Mapped Yes/No to 1/0 (Churn_Binary)
Integer as flag	SeniorCitizen	Mapped 0/1 to No/Yes for readability
Duplicate rows	All columns	Checked — 0 duplicates found

Feature categories

Demographics (4): gender, SeniorCitizen, Partner, Dependents. Account info (3): tenure, Contract, PaymentMethod. Services (8): PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV/Movies. Revenue (2): MonthlyCharges, TotalCharges. Target (1): Churn.

3. Key Findings from EDA & SQL Analysis

3.1 Contract type is the strongest churn predictor

Contract Type	Total Customers	Churned	Churn Rate	Risk Level
Month-to-month	3,875	1,655	42.7%	HIGH
One year	1,473	166	11.3%	MEDIUM
Two year	1,695	48	2.8%	LOW

Month-to-month customers churn at 15x the rate of two-year contract customers. This is the single highest-leverage segment for retention investment.

3.2 Tenure — the danger zone is the first 12 months

Segment	Customers	Churned	Churn Rate	Avg Monthly Charge
Early (0-12 months)	2,175	1,037	47.7%	\$65.40
Growing (13-24m)	1,112	306	27.5%	\$68.20
Mature (25-48m)	1,691	332	19.6%	\$63.80
Loyal (49+ months)	2,065	194	9.4%	\$60.10

Nearly 50% of all churn happens within the first year. The average churned customer has tenure of 18 months vs 38 months for retained customers.

3.3 Internet service & payment method

Internet Service	Churn Rate		Payment Method	Churn Rate
Fiber optic	41.9%		Electronic check	45.3%
DSL	19.0%		Mailed check	19.1%
No internet	7.4%		Bank transfer (auto)	16.7%
			Credit card (auto)	15.2%

4. Machine Learning Model Performance

Two gradient boosting models were trained on 80% of the data (5,634 rows) and evaluated on a held-out 20% test set (1,409 rows). Class imbalance (26.5% churn) was handled via `scale_pos_weight` in XGBoost and `is_unbalance=True` in LightGBM. Performance was confirmed with 5-fold stratified cross-validation.

Model	Test AUC-ROC	5-Fold CV Mean	CV Std Dev	Winner
XGBoost	0.843	0.843	±0.008	
LightGBM	0.848	0.848	±0.007	YES

Classification report — LightGBM (test set)

Class	Precision	Recall	F1-Score	Support
Retained	0.87	0.93	0.90	1,036
Churned	0.74	0.59	0.66	373
Weighted avg	0.84	0.84	0.84	1,409

SHAP feature importance — top 5 drivers

Rank	Feature	Direction	Business meaning
1	tenure	Low tenure = high churn risk	First-year customers at highest risk
2	Contract	Month-to-month = high churn risk	Lock-in dramatically reduces churn
3	MonthlyCharges	Higher charge = higher risk	Price sensitivity drives exit decisions
4	InternetService	Fiber optic = high churn risk	Product-market fit issue for fiber
5	TotalCharges	Low total = high churn risk	Early customers, not yet invested

5. Business Recommendations

The following three recommendations are ranked by expected retention impact and ease of implementation.

1 Convert month-to-month customers to annual contracts

Month-to-month customers churn at 43% vs 3% for two-year customers. A targeted offer timed at month 3-6 — before the main churn spike — could include a discounted annual rate, a feature unlock, or a loyalty reward. Even converting 10% of month-to-month customers to annual contracts would reduce overall churn rate by approximately 4 percentage points.

2 Launch a 90-day early-tenure onboarding program

Nearly 50% of all churn occurs in the first 12 months, with the steepest drop in month 1-5. A structured onboarding sequence — welcome call at day 3, feature tutorial at day 7, check-in at day 30, and a satisfaction survey at day 60 — targets the highest-risk window. Fiber optic subscribers should receive a dedicated track given their 42% churn rate.

3 Incentivize auto-pay enrollment for electronic check users

Electronic check customers churn at 45% vs 15-17% for auto-pay customers. The causality may be behavioral — customers who set up auto-pay are more committed — but either way, nudging electronic check payers toward auto-pay via a one-month discount or bill credit is low-cost and targets the highest-churn payment segment directly.

6. Project Summary

Phase	Focus	Key Output
1	Data cleaning & preparation	7,043 rows cleaned, nulls handled, types fixed
2	Exploratory data analysis	7 charts, churn drivers identified
3	SQL analysis	10 queries, revenue at risk quantified
4	ML modeling + SHAP	LightGBM AUC 0.848, top 5 drivers explained
5	Power BI dashboard	3-page interactive dashboard, 10 DAX measures
6	README + report	GitHub documentation, PDF report

Dataset: Telco Customer Churn (IBM Watson Analytics / Kaggle). All analysis performed independently for portfolio purposes.